

```

import time

n = int(input("Enter number of matrices: "))

print("\nEnter dimensions:")
print("For n matrices, enter n+1 dimensions.")
print("Example: For 10x20, 20x30, 30x40 enter: 10 20 30 40")

p = list(map(int, input("Enter dimensions: ").split()))

if len(p) != n + 1:
    print("Invalid input! You must enter", n + 1, "dimensions.")
    exit()

start_time = time.perf_counter()

m = [[0 for _ in range(n + 1)] for _ in range(n + 1)]

for length in range(2, n + 1):
    for i in range(1, n - length + 2):
        j = i + length - 1
        m[i][j] = float('inf')

        for k in range(i, j):
            cost = (
                m[i][k]
                + m[k + 1][j]
                + p[i - 1] * p[k] * p[j]
            )

            if cost < m[i][j]:
                m[i][j] = cost

end_time = time.perf_counter()

print("\nMinimum number of scalar multiplications:",
      m[1][n])

print("Execution time:",
      end_time - start_time, "seconds")

```

Enter number of matrices: 8

Enter dimensions:

For n matrices, enter n+1 dimensions.

Example: For 10x20, 20x30, 30x40 enter: 10 20 30 40

Enter dimensions: 4 5 2 6 1 7 3 9 5

Minimum number of scalar multiplications: 155

Execution time: 0.00043929299999945215 seconds